

# Notes: Kalouptside (AER, 2014)

## Time to Build and Fluctuations in Bulk Shipping

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# 1 Big picture

- **Question.** How much do construction lags, especially cyclical time to build, matter for investment, prices, profits, and consumer surplus in a volatile durable-goods industry?
- **Setting.** The paper studies the world Handysize bulk shipping market, where shipowners order vessels from shipyards, operate them for many years, trade them in a secondhand market, and scrap them at the end of life.
- **Core friction.** Ships take time to build, and construction delays are longer when industry investment is high because shipyards become congested.
- **Core insight.** Time to build smooths entry and dampens short-run supply responses. It raises freight prices and consumer costs, but it protects incumbent shipowners by limiting rapid entry after good demand shocks.
- **Estimation innovation.** Secondhand ship prices are used as direct observations of ship value functions. This lets the paper estimate dynamic value functions nonparametrically before backing out entry and exit primitives.
- **Main counterfactual.** Moving from endogenous time to build to constant or no time to build lowers shipping prices, raises consumer surplus, and greatly increases the level and volatility of investment.
- **Economic takeaway.** Time to build is not a small technological nuisance. It is a quantitatively important source of cycles, delayed adjustment, and welfare-relevant price volatility in bulk shipping.

# 2 Incremental contribution of the paper

1. **A structural empirical model of time to build.** The paper brings time-to-build frictions from macro and investment theory into a structurally estimated dynamic industry model.
2. **Endogenous rather than fixed construction lags.** Instead of treating construction time as a constant lag, the model allows delivery delays to lengthen in booms because shipyards have capacity constraints and order queues.
3. **A new use of secondhand asset prices.** The key methodological contribution is to use resale prices as value-function observations. Most dynamic-game estimation starts from payoffs and policies to infer value functions; this paper reverses the direction.
4. **Quantifying the welfare cost of sluggish capital adjustment.** The counterfactuals translate time-to-build frictions into prices, consumer surplus, profits, and investment volatility.
5. **Empirical link between micro asset markets and macro fluctuations.** The shipping industry provides a clean setting where asset resale prices, entry, exit, and demand shocks can be connected in one dynamic model.
6. **Policy and industry relevance.** The results matter for understanding capacity cycles in durable capital industries, not only for shipping. Similar time-to-build dynamics can arise in aircraft, energy, real estate, and industrial capacity.

# 3 Data and measurement

## 3.1 Industry and sample

The focus is on Handysize bulk carriers. The industry is made of many small shipowning firms, many of which own only a few ships. The paper treats a firm as a ship and models entry, aging, operation, resale, and exit. The empirical sample uses 1,838 secondhand sales from August 1998 through June 2010, voyage-contract data, fleet composition, entry and exit data, shipbuilding prices, and observed delivery lags.

**Interpretation:** The Handysize sector is useful because it is close to atomistic competition, has an active secondhand market, and has rich data on vessel age, size, resale prices, orders, exits, and prices.

### 3.2 Demand state variable

The demand state variable  $d_t$  is constructed from freight-market information rather than imposed as a single observable proxy. It is estimated from an inverse demand curve and then modeled as a persistent process with fat-tailed shocks:

$$d_t = c + \rho d_{t-1} + \epsilon_t.$$

The innovation distribution is a continuous mixture of normals, which permits large shocks like the 2008 crisis.

**Interpretation:** The persistent demand state is essential because entry decisions are forward-looking. If demand is high today but expected to mean-revert quickly, the value of ordering a ship differs from a case in which the shock is persistent.

### 3.3 Secondhand prices as value functions

Under homogeneous shipowners and a large number of potential entrants, the resale price of a ship of age  $j$  in state  $x = (S, B, d)$  equals its value function plus transaction noise:

$$p_{j,t}^{SH} = V_j(S_t, B_t, d_t) + \epsilon_{j,t}.$$

This is the paper’s main measurement bridge from data to dynamic values.

**Interpretation:** The secondhand market is an outside option for operating a ship. If a shipowner is indifferent between selling and keeping the vessel, the resale price reveals continuation value.

### 3.4 Time to build

Time to build is estimated from delivery lags for newly ordered vessels and is allowed to depend on state variables, especially the orderbook or entry intensity. The mean observed time to build rises sharply in the boom before 2008.

**Interpretation:** This is the central dynamic friction. A ship ordered in a boom may not be delivered until conditions have changed, so investment decisions embed substantial demand risk.

### 3.5 Entry, exit, and scrap values

Exit occurs when a ship is scrapped. Scrap value is estimated from observed demolition values. Entry costs are recovered from a free-entry condition using the age-zero value function and the estimated transition matrix. Exit costs are inferred from continuation values and scrap values.

**Interpretation:** Entry and exit are asymmetric because ships cannot be unscrapped and entry requires long construction delays. This creates hysteresis and slow industry adjustment.

## 4 Model and empirical specification

### 4.1 State variables and decisions

The industry state is summarized by

$$x_t = (S_t, B_t, d_t),$$

where  $S_t$  is the operating fleet,  $B_t$  is the orderbook under construction, and  $d_t$  is the demand state. A ship of age  $j$  receives current profits, ages, and chooses whether to exit. Potential entrants decide whether to order new ships, which are delivered after a state-dependent time-to-build lag.

### 4.2 Demand and profits

Freight demand determines the shipping price. Operating profits for a ship depend on the freight rate, operating costs, age, and demand state. The model estimates profits nonparametrically for older ships and parametrically for younger ships where data are thinner.

### 4.3 Dynamic value function

A ship of age  $j$  has value

$$V_j(x) = \pi_j(x) + \beta E[\max\{V_{j+1}(x'), \phi_j\} \mid x],$$

where  $\phi_j$  is the scrap option. The value incorporates current operating profits, expected future industry states, and the endogenous exit option.

### 4.4 Entry condition

For a new ship, the free entry condition equates expected discounted value upon delivery to the entry cost:

$$\beta^{T(x)} E[V_0(x_{t+T(x)}) \mid x_t] = \kappa(x_t),$$

where  $T(x)$  is the time to build.

**Interpretation:** A higher demand state raises the value of entry, but also lengthens construction delays by increasing the orderbook. This makes investment less responsive than in a model with instantaneous entry.

### 4.5 Estimation strategy

The empirical strategy proceeds in steps:

1. Estimate the demand curve and construct the demand state  $d_t$ .
2. Estimate the transition process for  $d_t$  with nonnormal innovations.
3. Use secondhand prices to estimate value functions by age and state.
4. Estimate time to build from observed delivery lags.
5. Recover profits, scrap values, exit costs, and entry costs from value functions and policy restrictions.
6. Run counterfactuals with endogenous, constant, and no time to build.

## 5 Identification and empirical design

### 5.1 What identifies demand and demand shocks

Demand is identified from freight-market data using the inverse demand relationship and instruments. The resulting demand state variable summarizes global transportation demand, especially the boom driven by China's raw material imports.

### 5.2 What identifies value functions

Secondhand ship sales identify value functions because the resale price is the market value of a ship with a given age in a given state. The identification requires that sellers and buyers share the same valuation and that ship quality is sufficiently observable in secondhand transactions.

### 5.3 What identifies time to build

Observed order and delivery dates identify delivery lags. Variation over time in the orderbook and entry intensity identifies how time to build changes with market conditions.

### 5.4 What identifies entry and exit costs

Entry costs are identified from the free-entry condition. Scrap values and exit behavior identify continuation values and exit costs. The model uses the fact that scrapping is irreversible and that secondhand values reveal the outside option of operation.

### 5.5 Main assumptions and threats

- **Homogeneous shipowners:** resale prices reveal common values rather than private valuations.
- **Large number of potential entrants:** free entry pins down entry costs.
- **Observed ship quality:** unobserved quality should not dominate resale prices.
- **Exogenous demand process:** freight demand shocks are treated as exogenous to Handysize entry decisions.
- **Approximated competition:** the empirical strategy relies on an industry with many small firms; counterfactuals compare with perfect competition as a special case.

**Interpretation:** The design is structural rather than quasi-experimental. Credibility rests on the fit of prices, entry, exit, time to build, and the model's ability to reproduce observed industry dynamics.

## 6 Tables and figures

### 6.1 Figure 1: Fleet and Fleet Shares of Shipowning Firms; Figure 2: China's Iron Ore and Coal Imports

**Read:** Figure 1 shows a highly fragmented industry with many small shipowners. Figure 2 shows the sharp rise in China's iron ore and coal imports before the financial crisis.

**Economic message:** The first figure justifies treating the industry as atomistic or close to competitive. The second figure motivates the demand boom that stresses the time-to-build friction.

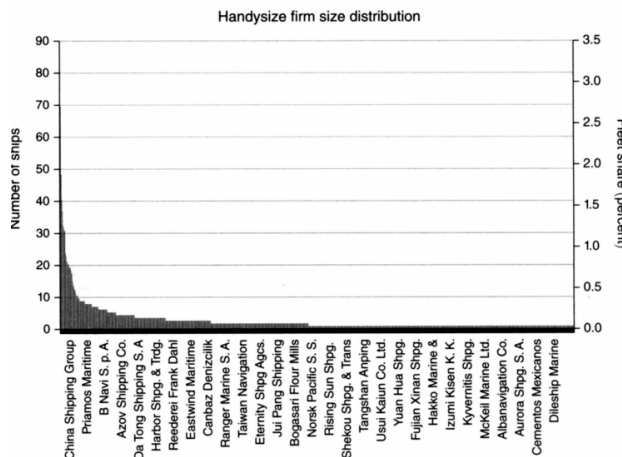


FIGURE 1. FLEET AND FLEET SHARES OF SHIPOWNING FIRMS

**Figure 1: Fleet and Fleet Shares of Shipowning Firms.** A long right tail indicates that most firms are small, so a ship-level model is empirically plausible.

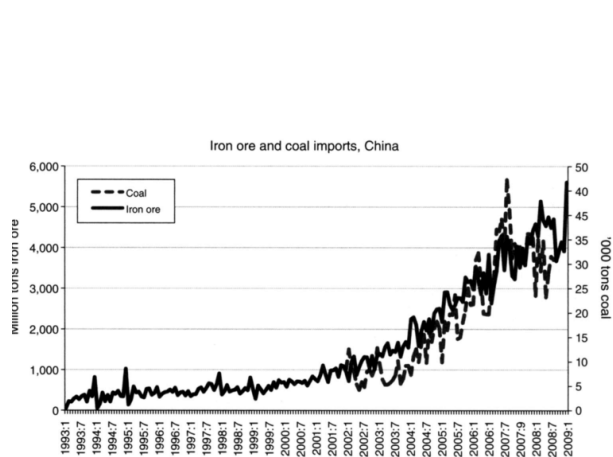


FIGURE 2. CHINA'S IRON ORE AND COAL IMPORTS

**Figure 2: China's Iron Ore and Coal Imports.** Chinese commodity demand rose sharply before 2008, creating a large positive demand shock for bulk shipping.

## 6.2 Figure 3: The Baltic Dry Index; Table 1: Secondhand Sales Summary Statistics

**Read:** Figure 3 shows extreme freight price volatility, especially the 2007-2008 spike and subsequent crash. Table 1 reports the secondhand ship sales sample.

**Economic message:** Freight-rate volatility creates the investment-under-uncertainty problem. Table 1 shows that resale prices and ship ages vary enough to identify value functions.



FIGURE 3. THE BALTIC DRY INDEX

otes: Daily index based on weighted average of rates on 20 representative bulk routes. Compiled by the Baltic xchange. 1/11/1999 = 1,334.

**Figure 3: The Baltic Dry Index.** The volatility of freight rates is the demand-side force that interacts with slow supply adjustment.

TABLE 1—SECONDHAND SALES SUMMARY STATISTICS

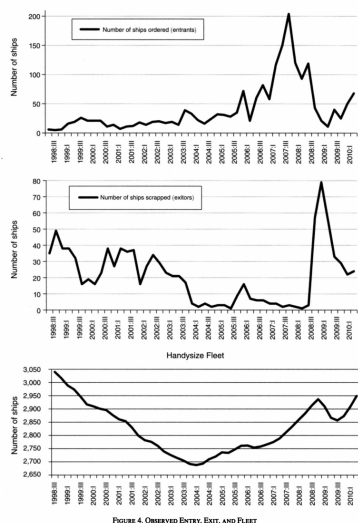
|                             | Mean   | SD    |
|-----------------------------|--------|-------|
| Price in million US dollars | 9.97   | 8.54  |
| Age in years                | 18.7   | 7.6   |
| Size in DWT                 | 27,876 | 6,718 |
| Sales in a quarter          | 41.85  | 15.3  |
| Observations                | 1,838  |       |

**Table 1: Secondhand Sales Summary Statistics.** The secondhand sales data supply direct observations of ship values by age, size, and market state.

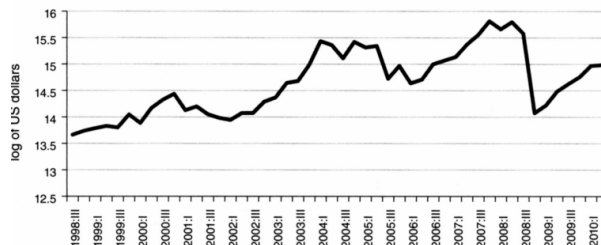
## 6.3 Figure 4: Observed Entry, Exit, and Fleet; Figure 5: Demand State Variable

**Read:** Figure 4 shows entry, exit, and fleet dynamics. Figure 5 shows the estimated demand state variable.

**Economic message:** Entry surges during booms but cannot immediately add capacity. Exit spikes in the crisis, while the fleet adjusts with a lag. The demand state summarizes the market environment driving those decisions.



**Figure 4: Observed Entry, Exit, and Fleet.** Entry and exit are highly cyclical, while the fleet moves slowly.



**FIGURE 5. DEMAND STATE VARIABLE**

**Figure 5: Demand State Variable.** The latent demand state rises during the boom and falls sharply with the crisis.

## 6.4 Table 2: Demand Transition Parameters; Figures 6 and 7: Value Functions

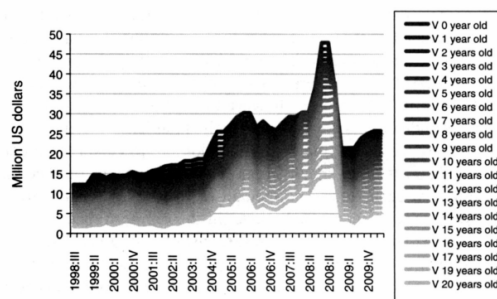
**Read:** Table 2 reports the AR(1) demand process and fat-tail parameters. Figures 6 and 7 show estimated value functions by ship age and confidence intervals.

**Economic message:** Demand is persistent, and the value of ships rises sharply in booms. Newer ships have larger values because they have more future operating life.

|                | $c$     | $\rho$  | $a$     | $\gamma$ |
|----------------|---------|---------|---------|----------|
| Parameter      | 0.7483  | 0.9542  | 1.575   | 25.356   |
| Standard error | (0.698) | (0.048) | (0.044) | (7.784)  |

*Note:* Standard errors based on 500 bootstrap samples.

**Table 2: Demand Transition Parameters.** The high persistence parameter shows that demand shocks affect long-run investment values.



**FIGURE 6. VALUE FUNCTIONS OF YEARLY AGES**

**Figure 6: Value Functions of Yearly Ages.** Ship values rise with demand and decline with age.



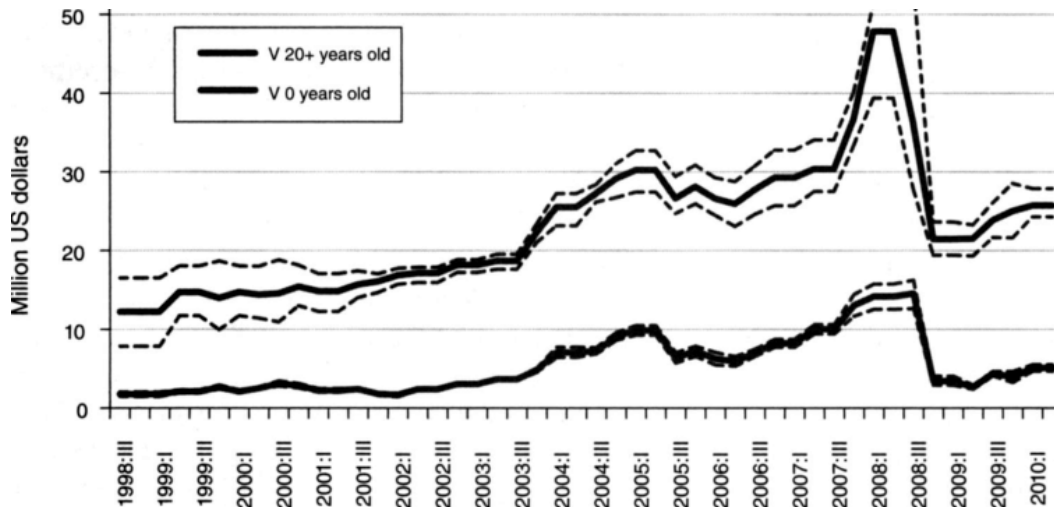


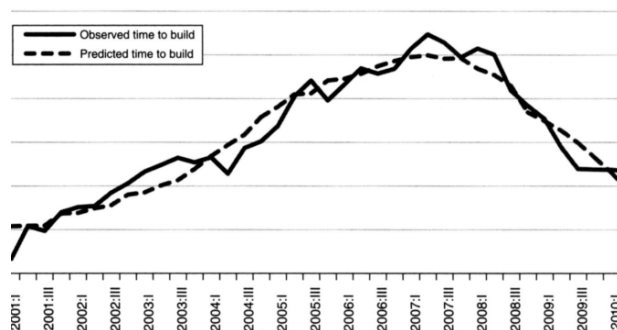
FIGURE 7. VALUE FUNCTIONS AND CONFIDENCE INTERVALS

**Figure 7: Value Functions and Confidence Intervals.** Confidence intervals show the precision of the age-zero and old-ship value estimates.

## 6.5 Figure 8: Average Observed and Estimated Time to Build; Figure 9: Distribution and Density of Scrap Values

**Read:** Figure 8 compares observed and predicted time to build. Figure 9 reports scrap-value distributions and densities.

**Economic message:** Time to build becomes much longer in booms. Scrap values provide the outside option for exit and quantify irreversibility on the exit margin.



**Figure 8: Average Observed and Estimated Time to Build.** The model captures the boom-time lengthening of delivery lags.

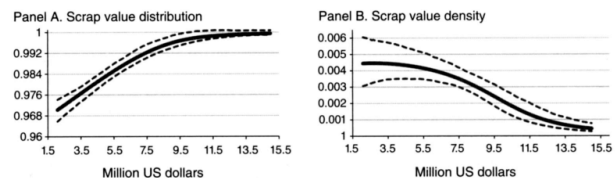


FIGURE 9. DISTRIBUTION AND DENSITY OF SCRAP VALUES

Note: Dashed lines denote 95 percent bootstrap confidence intervals.

**Figure 9: Distribution and Density of Scrap Values.** Scrap-value estimates discipline the exit option for old ships.

## 6.6 Figures 10 and 11: Estimated Profits and Earnings Index

**Read:** Figure 10 reports estimated profits for old ships. Figure 11 compares the weighted average of nonparametric profits with the Clarkson earnings index.

**Economic message:** The profit estimates track broker-based earnings, validating the nonparametric profit recovery used in the model.

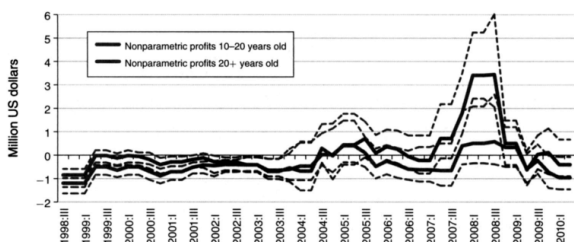


FIGURE 10. ESTIMATED PROFITS

Notes: Average across age 10–20 years and 20+ years old. 0.95 bootstrap confidence intervals.

**Figure 10: Estimated Profits.** Profits spike in the 2007–2008 boom and collapse in the crisis.

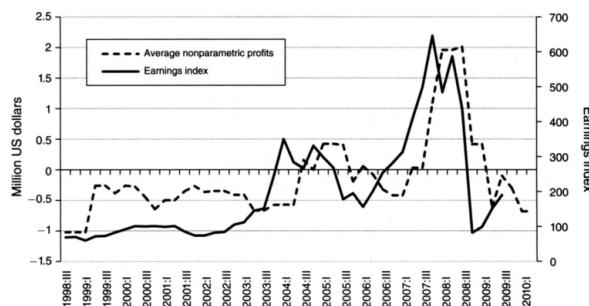


FIGURE 11. WEIGHTED AVERAGE OF NONPARAMETRIC PROFITS AND EARNINGS INDEX

Note: Index based on 45000DWT Handymax, based on freight rates and fuel/port costs for representative routes.

**Figure 11: Weighted Average of Nonparametric Profits and Earnings Index.** The close co-movement supports the model's profit estimates.

## 6.7 Figure 12: Estimated Entry Costs and Average Shipbuilding Price; Table 3: Profit Parameters

**Read:** Figure 12 compares estimated entry costs with the average shipbuilding price. Table 3 reports profit-function parameters.

**Economic message:** Entry costs move with shipbuilding prices but also include nonprice costs such as contracting, negotiation, and supervision.

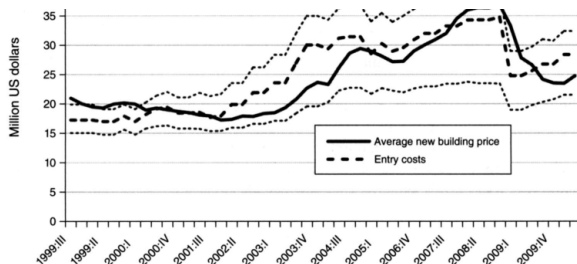


FIGURE 12. ESTIMATED ENTRY COSTS AND AVERAGE SHIPBUILDING PRICE

Note: Dashed lines denote 95 percent bootstrap confidence intervals.

**Figure 12: Estimated Entry Costs and Average Shipbuilding Price.** Entry costs rise sharply in the boom.

TABLE 3—PROFIT PARAMETERS

|                 | $c_{0-39}$ | $c_{40-79}$ | $c_{80+}$ | $F_{0-39}$ | $F_{40-79}$ | $F_R$     |
|-----------------|------------|-------------|-----------|------------|-------------|-----------|
| Parameters      | 300,387    | 466,683     | 651,456   | 492,307    | 648,282     | 821,456   |
| Standard errors | (103,871)  | (156,016)   | (216,789) | (347,982)  | (347,977)   | (347,977) |

Standard errors based on 500 bootstrap samples.

$\in \{0-39, 40-79, 80+\}$  representing the three age groups. Assuming that profits change by age decade ensures that profits depend on the specified variables.

**Table 3: Profit Parameters.** Parameters summarize how profits vary with demand, age, and the fleet state.

## 6.8 Figure 13: Results of Perfectly Competitive Model

**Read:** Figure 13 compares model outcomes under a perfectly competitive benchmark.

**Economic message:** This verifies that the model's counterfactual mechanism is not driven solely by market power assumptions. The results remain close to the main baseline.

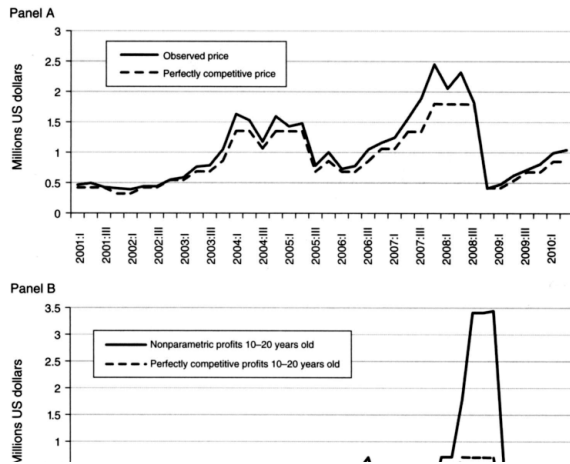


Figure 13, part 1: Perfectly competitive model. Entry and price dynamics remain similar to the main model.

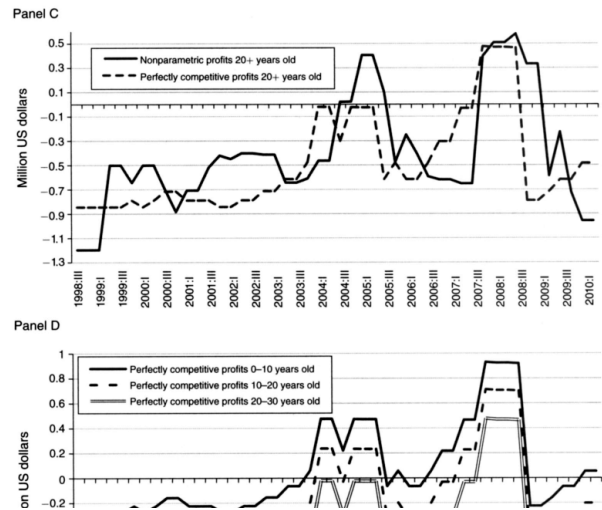


Figure 13, part 2: Perfectly competitive model continued. Continued outcomes support the approximation of a competitive market.

## 6.9 Figures 14 and 15: Entry and Exit under Endogenous, Constant, and No Time to Build

**Read:** Figure 14 shows entry under alternative time-to-build assumptions. Figure 15 shows exit.

**Economic message:** Removing or fixing time to build changes the timing and volatility of entry. Exit responds less violently when endogenous time to build dampens entry booms.

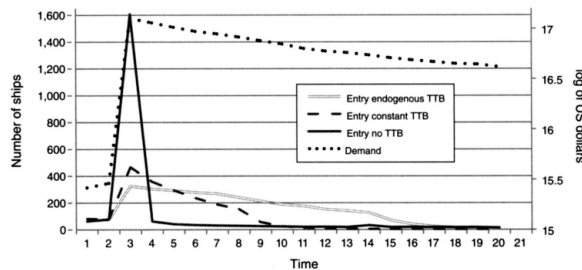


FIGURE 14. ENTRY UNDER ENDOGENOUS, CONSTANT, AND NO TIME TO BUILD (TTB)

Note: Demand depicted on right axis.

Figure 14: Entry under Endogenous, Constant, and No Time to Build. No time to build produces a sharper entry response.

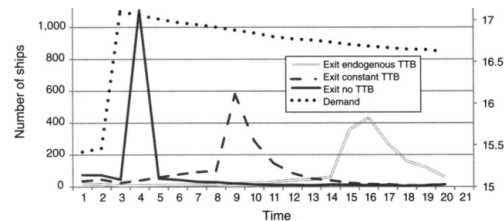


FIGURE 15. EXIT UNDER ENDOGENOUS, CONSTANT, AND NO TIME TO BUILD (TTB)

Note: Demand depicted on right axis.

Figure 15: Exit under Endogenous, Constant, and No Time to Build. Exit spikes when prior entry booms create excess capacity.

## 6.10 Figures 16 and 17: Price and Consumer Surplus under Alternative Time to Build

**Read:** Figure 16 shows prices, and Figure 17 shows consumer surplus.

**Economic message:** Shorter time to build lowers prices and raises consumer surplus because supply expands faster after a demand shock.

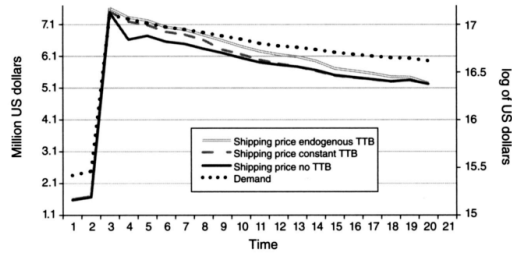


FIGURE 16. PRICE UNDER ENDOGENOUS, CONSTANT, AND NO TIME TO BUILD (TTB)

Note: Demand depicted on right axis.

**Figure 16: Price under Endogenous, Constant, and No Time to Build.** Prices are highest when supply adjustment is slowest.

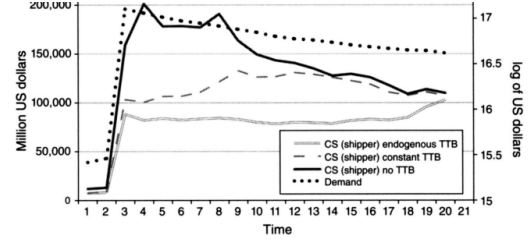


FIGURE 17. CONSUMER SURPLUS (*i.e.*, surplus of freight customers) UNDER ENDOGENOUS, CONSTANT, AND NO TIME TO BUILD (TTB)

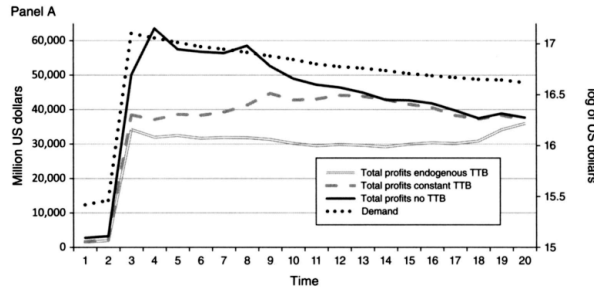
Note: Demand depicted on right axis.

**Figure 17: Consumer Surplus under Alternative Time to Build.** Consumers gain when time to build is shorter because freight prices fall.

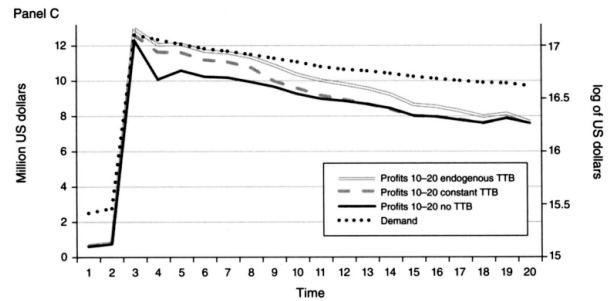
## 6.11 Figure 18: Total and per-Ship Profits for All Age Groups; Table 4: Ratio of Constant to Endogenous TTB

**Read:** Figure 18 compares total and per-ship profits by age group. Table 4 summarizes ratios of constant versus endogenous time-to-build outcomes.

**Economic message:** Time to build protects incumbents from immediate entry, raising per-ship profits. But it also reduces consumer surplus.



**Figure 18, part 1: Total and per-ship profits.** Profits differ by age and respond to entry constraints.



**Figure 18, part 2: Total and per-ship profits continued.** The delayed-entry effect is visible across age groups.

TABLE 4—COMPARISON OF RATIO OF CONSTANT TO ENDOGENOUS TIME TO BUILD (CT/ET)

|                             | CT/ET | NT/ET |
|-----------------------------|-------|-------|
| Fleet level                 | 1.008 | 1.16  |
| Price level                 | 0.999 | 0.97  |
| Entry level                 | 1.016 | 1.165 |
| Fleet variance              | 1.44  | 2.06  |
| Price variance              | 0.98  | 0.86  |
| Entry variance              | 2.11  | 7.64  |
| Fleet and demand covariance | 1.19  | 2.42  |
| Price and demand covariance | 0.995 | 0.945 |
| Entry and demand covariance | 1.19  | 1.763 |

Table 4: Comparison of Ratio of Constant to Endogenous Time to Build. The table quantifies how counterfactual time-to-build assumptions change prices, profits, and welfare measures.

## 6.12 Figures 19, 20, and 21: Long-run Simulations

**Read:** Figure 19 shows one simulated industry path. Figures 20 and 21 report average fleet and entry from long-run simulations.

**Economic message:** Endogenous time to build generates persistent cyclical dynamics and investment echo effects after shocks.

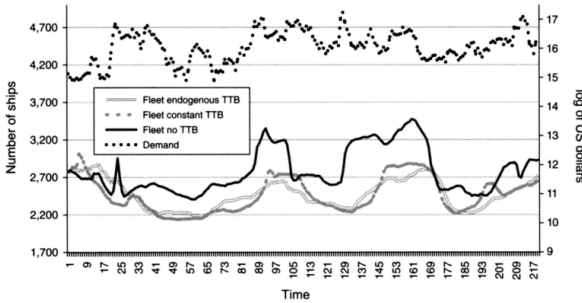


FIGURE 19. A SAMPLE INDUSTRY PATH

Note: Demand depicted on right axis.

**Figure 19: A Sample Industry Path.** A demand shock triggers delayed entry and persistent fluctuations.

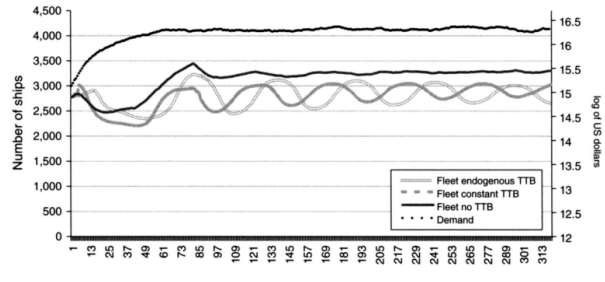


FIGURE 20. AVERAGE FLEET FROM LONG-RUN SIMULATIONS

Note: Demand depicted on right axis.

**Figure 20: Average Fleet from Long-Run Simulations.** Fleet adjustment is slow and hump-shaped.

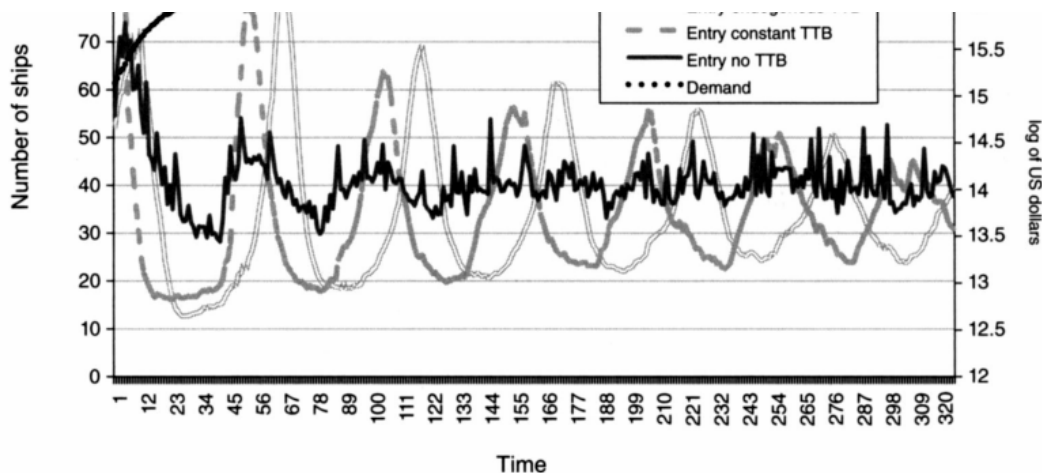


FIGURE 21. AVERAGE ENTRY FROM LONG-RUN SIMULATIONS

*Note:* Demand depicted on right axis.

Figure 21: Average Entry from Long-Run Simulations. Entry exhibits delayed responses and fading echo effects.

### 6.13 Appendix Tables 5-8: Demand, Time to Build, Entry, and Exit Regressions

**Read:** These appendix tables support the construction of demand, time-to-build, entry, and exit transition components.

**Economic message:** The appendix tables provide reduced-form evidence that underlies the model primitives used in the structural estimation and counterfactuals.

TABLE 5—INVERSE DEMAND CURVE FOR FREIGHT TRANSPORT: IV REGRESSION RESULTS

|                 | First stage, dep. variable $Q_i$ |           | Second stage, dep. variable $P_i$ |           |
|-----------------|----------------------------------|-----------|-----------------------------------|-----------|
|                 | Parameter                        | SE        | Parameter                         | SE        |
| const           | 2.01                             | (20)      | -7.601                            | (23.8)    |
| WIP             | -5.05                            | (3.4)*    | 9.501                             | (4.51)**  |
| agr raw mat $P$ | 1.291                            | (0.97)*   | 2.969                             | (1.32)**  |
| mineral $P$     | 0.394                            | (0.57)    | -1.658                            | (0.565)** |
| food $P$        | -0.548                           | (0.715)   | -0.346                            | (0.702)   |
| China steel     | 0.365                            | (0.716)   | 1.534                             | (0.592)** |
| Handymax        | -2.03                            | (2.12)    | -4.705                            | (1.324)** |
| fleet           | 0.0013                           | (0.0014)  |                                   | (0.597)   |
| mean age fl     | 0.287                            | (0.150)** |                                   |           |
| std age fl      | 0.5823                           | (0.335)** |                                   |           |

Table 5: Inverse Demand Curve for Freight Transport. IV estimates used to construct the demand state.

\*\*Significant at the 5 percent level.  
\*Significant at the 10 percent level.

TABLE 6—TIME TO BUILD REGRESSION ESTIMATES

|                 | Constant | $S^1$     | $S^2$     | $S^3$     | $B$          | $d$     |
|-----------------|----------|-----------|-----------|-----------|--------------|---------|
| Parameters      | 2.536    | -0.00082  | -0.00063  | 0.00011   | 1.93e - 005  | 0.0303  |
| Standard errors | (1.266)  | (0.00058) | (0.00036) | (0.00036) | (8.3e - 005) | (0.019) |

*Notes:* Standard errors based on 500 bootstrap samples. Coefficients joint significant at the 0.01 level.

Table 6: Time to Build Regression Estimates. Delivery lags rise with order intensity and market state.

TABLE 7—ENTRY AND EXIT REGRESSION ESTIMATES

|                 | Constant | $S^1$    | $S^2$    | $S^3$    | $B$       | $B^*$              | $d$      |
|-----------------|----------|----------|----------|----------|-----------|--------------------|----------|
| <i>Entry</i>    |          |          |          |          |           |                    |          |
| Parameters      | -3.68    | -0.00134 | -0.00019 | -0.00014 | 0.0069    | $-5.37e - 006$     | 0.486    |
| Standard errors | (7.99)   | (0.0049) | (0.0019) | (0.0018) | (0.0024)* | ( $2.12e - 006$ )* | (0.22)*  |
| <i>Exit</i>     |          |          |          |          |           |                    |          |
| Parameters      | 22.68    | 0.0040   | 0.000178 | 0.00050  | -0.0042   | $4.141e - 006$     | -1.585   |
| Standard errors | (10.15)* | (0.0065) | (0.0023) | (0.0023) | (0.0036)  | ( $3.09e - 006$ )  | (0.34)** |

Note: Standard errors based on 500 bootstrap samples.

\*\*Significant at the 5 percent level.

\*Significant at the 10 percent level.

**Table 7: Entry and Exit Regression Estimates.**  
Entry and exit transition estimates.

TABLE 8—ENTRY AND EXIT REGRESSION ESTIMATES

|                 | Constant | $S^1$      | $S^2$     | $S^3$    | $d$       |
|-----------------|----------|------------|-----------|----------|-----------|
| <i>Entry</i>    |          |            |           |          |           |
| Parameters      | -8.425   | -0.0024    | -0.00045  |          | 0.934     |
| Standard errors | (4.90)   | (0.0025)   | (0.00075) |          | (0.244)** |
| <i>Exit</i>     |          |            |           |          |           |
| Parameters      | 22.728   | 0.0073     | 0.00093   | 0.00104  | -1.859    |
| Standard errors | (4.89)** | (0.0016)** | (0.00092) | (0.0008) | (0.242)** |

Note: Standard errors based on 500 bootstrap samples.

\*\*Significant at the 5 percent level.

**Table 8: Entry and Exit Regression Estimates.**  
Alternative specification for entry and exit transitions.

## 7 Short summary

- The paper studies bulk shipping as a setting with large demand shocks and slow capital adjustment.
- Time to build is endogenous: ship delivery lags lengthen in investment booms.
- Secondhand prices identify value functions directly.
- Persistent demand shocks and fat-tailed innovations are needed to match the shipping boom and crash.
- Entry costs and time-to-build delays make investment sluggish.
- Removing time to build lowers prices and raises consumer surplus but increases investment volatility.
- Incumbent shipowners benefit from slow entry because high demand cannot immediately be met with new capacity.
- The paper’s method shows how asset resale markets can discipline dynamic structural models.

## 8 Conclusion

### 8.1 Core conclusion

Kalouptsidei shows that time to build is a quantitatively important source of fluctuations in bulk shipping. Construction lags delay the supply response to demand booms, raise freight prices, and create persistent investment cycles.

### 8.2 Economic intuition

When demand for shipping rises, shipowners want to order new vessels. But ships take years to build, and shipyard congestion makes delays longer exactly when demand is high. Supply therefore arrives late, sometimes after the boom has ended. This delayed adjustment creates cycles in prices, entry, exit, profits, and consumer surplus.

### 8.3 Why the paper matters

The paper makes a methodological contribution by using secondhand ship prices as value-function observations. This allows the model to be estimated without first imposing a fully parametric profit structure. It also provides one of the clearest empirical measurements of time-to-build frictions in a durable capital industry.

## 8.4 Policy and welfare implications

Consumers of shipping services would benefit from faster capacity adjustment because freight prices would be lower and consumer surplus higher. Incumbent shipowners, however, benefit from time to build because it slows entry and keeps prices high in booms. The welfare consequences therefore differ across freight customers, incumbents, entrants, and shipyards.

## 8.5 Limitations

The model relies on homogeneous shipowners and the interpretation of resale prices as common value functions. It also treats demand shocks as exogenous and focuses on one ship size segment. These assumptions are useful for identification but may limit external validity.

## 8.6 Takeaway

The main takeaway is that time to build is a powerful adjustment friction. In industries where capacity takes years to produce, demand shocks can generate long-lasting cycles, even without large strategic distortions or irrational expectations.